Fiscal Year:	FY 2011	**Task Last Updated:**	FY 02/10/2012
PI Name:	McKinley, Gareth Ph.D.		
Project Title:	Shear History Extensional Rheology Experiment II (SHERE II)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Fluid Physics Rapid Turn Around Flight Opportunity
Start Date:	09/03/2009	**End Date:**	09/02/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 9/02/2012 per NSSC information (Ed., 2/25/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Haward, Simon (Massachusetts Institute of Technology) Jaishankar, Aditya (Massachusetts Institute of Technology)		
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Task Description:	The SHERE II experiment was operated on board the International Space Station (ISS) during Increment 27/28. SHERE II is an extension of the SHERE project, and both used the same hardware but different test fluid. The experiment is a glovebox investigation developed at NASA Glenn Research Center (GRC) and intended for operation in the Microgravity Science Glovebox (MSG) aboard the ISS. It is the first instrument capable of assessing the effects of a pre-shear history on the extensional properties of dilute polymer solutions during the extended range of time scales that can be accessed because of the absence of gravitational sagging. As a logical extension to the SHERE project, SHERE II looks to study the effect of torsional pre-shear on the extensional behavior of a different non-Newtonian fluid. We specifically chose to study dilute suspensions of rigid micro-particles in dilute polymer solutions (0.025 wt. %). The suspending particles used are rigid inert poly(methyl methacyrlate) (PMMA) spheres, 6 μm in diameter, and the polymeric matrix is narrow polydispersity high molecular weight polystyrene dissolved in oligomeric styrene, which is the same fluid used in SHERE. The SHERE II experiment will allow us to measure for the first time the effects of rotational pre-shear on the axial stretching of a polymeric fluid with small particles dispersed within it. The experiment also allows the study of the fluid's relaxation after the axial stretching, which is only possible in a microgravity environment. The results will provide new insights into the behavior of non-Newtonian fluids in industrial processes with both rotational and axial flows. The results of SHERE II will provide high quality experimental data to compare with recently developed theories on filler materials in plastics. Comparison of the SHERE (no particles) and SHERE II (particles) experiments is expected to be enlightening. The goal of the SHERE II experiment is to study the effects of a rotational pre-shear history on the transient extensional viscosity in a uniaxial stretching flow for dilute suspensions of inert, rigid micro-spheres (particles) suspended in a viscoelastic matrix fluid consisting of a dilute polymer solution. Constitutive equations for viscoelastic fluids such as dilute polymer or wormlike micelles solutions generally predict large changes in the extensional rheology of the material as the elongation rate is increased. However, the validity of these theories has not yet been confirmed due to the lack of experimental data obtained in extensional flows. Prior to this research, quantitative rheological measurements in shear-free flows were only possible for highly elastic materials such as polymer melts which could easily be elongated without sagging under a gravitational body force. The SHERE and SHERE II experiments allow us to obtain, for the first time, accurate measurements of the extensional viscosity for less elastic fluids such as polymer solutions, suspensions and liquid crystalline materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the rheological properties of highly viscoelastic suspensions may be of paramount importance for lunar in-situ resource utilization and for the future construction of a permanent lunar base. Indeed, the lunar soil, referred to as regolith, is composed of loose dust and broken rocks that could be used as a high volume fraction filler when mixed with a polymeric binder or matrix for building foundation or roadbed materials. The viscoelastic dilute suspension used for SHERE II thus serves as a highly idealized model system amenable to theoretical analysis which can shed light on future development of lunar regolith-based construction materials. Suspensions of particles in viscoelastic liquids are used in many earth processing operations: polymer melts with fillers, ceramic pastes, biomedical materials, food, cosmetics and detergents. The final properties of the suspensions are greatly determined by the shape, concentration and size of the filler. In particular, the fillers can range from nanoscopic to microscopic characteristic dimensions, which leads to very different types of flow behaviors, filler/matrix interactions and dynamics. The results of SHERE II will provide new insights into the behavior of this important class of non-Newtonian fluids in industrially relevant mixed shear and extensional flows. This information is particularly relevant for understanding the deformation and evolution of fluid threads in a wide variety of industries, including polymer processing, food and consumer product processing, biomedical diagnostics, and pressure-sensitive adhesives. This research has direct applications in many industrial processes such as fiber spinning, film coating, injection molding, extrusion, and blow molding. Additionally, shear free flows arise whenever liquid interfaces are present; for example in "containerless processing" operations. These rheological data obtained will enable designers of both space- and ground-based material processes to use improved constitutive models in numerical simulations of complex two- and three-dimensional fluid flows. This includes stability and breakup of jets, enhanced oil recovery, and turbulent drag reduction for advanced aircraft, boats, and submarines.
Task Progress:	The inert rigid PMMA spheres are dispersed in the polystyrenre/oligomeric styrene Boger fluid matrix at a mass fraction 0.04, which is large enough to give a measurable increase in the shear viscosity and extensional viscosity but small enough to remain within the bounds of the asymptotic theory. At larger volume fractions, the rheology of filled suspensions become increasingly nonlinear and may become shear-thickening or develop a yield stress (like toothpaste, which is a highly filled weakly viscoelastic suspension of silica). To make the present batch of suspension, 7.679 g of beads were mixed into 181.5 g of matrix to give a final suspension concentration of cbead = 4.06 wt. %. The volume fraction is equal to 0.035 (or 3.5 vol.%). In the PI's Laboratory, extensive rheological tests were performed on the prepared suspension to characterize its viscometric properties in comparison to the Newtonian oligomeric styrene solvent and the polystyrene Boger fluid. The viscosity of the suspension is slightly higher than that of the matrix fluid and it shows some shear-thickening at high shear rates. The relaxation time of the suspension is similar to that of the pure matrix fluid. Aboard the ISS, astronaut Mike Fossum commenced SHERE II operations with the viscoelastic suspension on GMT 206 during week 19 of Increment 27/28. After solving some initial teething problems with the instrument software, on GMT 215 several dry run tests of the SHERE apparatus were successfully completed by Mike Fossum, the results of which indicated that all components of the SHERE instrument were functioning correctly. On GMT 220 Mike Fossum successfully completed the first two test points of the test matrix and preliminary analysis of the data was positive and encouraging. Astronauts Satoshi Furukawa and Mike Fossum proceeded to complete the remaining 23 test points between GMT 223 and GMT 230. The full data analysis of all test points will be completed following the return to earth of the high-quality onboard video recordings.
Bibliography Type:	Description: (Last Updated: 09/04/2025)

Articles in Peer-reviewed Journals	Haward SJ, Ober TJ, Oliveira MSN, Alves MA, McKinley GH. "Extensional rheology and elastic instabilities of a wormlike micellar solution in a microfluidic cross-slot device." Soft Matter 2012;8(2):536-55. http://dx.doi.org/10.1039/C1SM06494K , Jan-2012
Articles in Peer-reviewed Journals	Haward SJ, McKinley GH. "Stagnation point flow of wormlike micellar solutions in a microfluidic cross-slot device: Effects of surfactant concentration and ionic environment." Physical Review E, In press. February 2012. , Feb-2012